

Advanced (Carolina Reaper)

- Q1.** What is the primary purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the median and interquartile range
(C) To compare the range of multiple datasets
(D) To identify the mode of a dataset
(E) To calculate the standard deviation
- Q2.** A student scored 85, 90, 78, 92, 88 on their quizzes. What is the interquartile range?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q3.** Which of the following is an outlier in the dataset 2, 4, 6, 8, 100?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 100
- Q4.** A box-and-whisker plot shows a median of 10 and an interquartile range of 5. What is the third quartile?
(A) 12
(B) 12.5
(C) 13
(D) 15
(E) 15.5
- Q5.** What does the length of the whisker represent in a box-and-whisker plot?
(A) The range of the dataset
(B) The interquartile range
(C) The median of the dataset
(D) The mode of the dataset
(E) The standard deviation
- Q6.** The prices of 5 video games are 0, 50, 60, 70, 80. What is the first quartile?
(A) 40
(B) 45
(C) 50
(D) 55
(E) 60
- Q7.** The times to complete a puzzle are 20, 25, 30, 35, 40 minutes. What is the interquartile range?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q8.** Which of the following datasets has the smallest interquartile range?
(A) 1, 2, 3, 4, 5
(B) 2, 4, 6, 8, 10
(C) 10, 20, 30, 40, 50
(D) 100, 200, 300, 400, 500
(E) 1, 1, 1, 1, 1

Open Ended Questions (FRQ)

- Q9.** Create a box-and-whisker plot for the dataset 12, 15, 18, 20, 22, 25, 30 and identify the median, first quartile, third quartile, and any outliers.
- Q10.** A company recorded the following daily sales: 200, 220, 250, 280, 300, 320, 350. Calculate the interquartile range and determine if there are any outliers. Justify your answer.

Chef's Tips

- Check for any calculations or graphing errors
- Ensure all answers are clearly labeled and easy to read
- Consider using technology to verify any calculations